CNS2025: Homework 15

Due: 2025-12-17 23:59

The trajectories shown in Slide 14 of the lecture follow the Hebbian weight dynamics

$$\tau_w \frac{d\mathbf{w}}{dt} = \mathbf{Q} \cdot \mathbf{w}$$

with saturation. Using the Oja rule of multiplicative normalization, the weights should follow

$$\begin{aligned}
\tau_w \frac{d\mathbf{w}}{dt} &= v\mathbf{u} - \alpha v^2 \mathbf{w} \\
&= (\mathbf{u} \cdot \mathbf{w})\mathbf{u} - \alpha (\mathbf{u} \cdot \mathbf{w})^2 \mathbf{w} \\
&= \mathbf{Q} \cdot \mathbf{w} - \alpha (\mathbf{w}^T \cdot \mathbf{Q} \cdot \mathbf{w})\mathbf{w}.
\end{aligned}$$

Find the code in `code15.ipynb` that produces the trajectories in Slide 14 and change it to the Oja rule. Make the same plot of trajectories with the Oja rule using $\alpha = 1$.
